

# GAME THEORY, Spring 2026

## Problem Set 1

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### 1 $n$ -Bidder Auction

Consider the first/second price auction environment we covered in class. Instead of 2 bidders, assume there are  $n$  bidders with value  $v_1, \dots, v_n$ .

- (a) Extend the first price auction to this case. As usual, if more than one bidder bid the same highest price, a winning bidder is randomly drawn from them with equal probabilities. (Use this tie breaking rule also for the next question).
- (b) Extend the second price auction to this case.

**Answer:** We write the induced normal-form game in the standard way.

- (a) **First-price auction.** Let the set of players be  $N = \{1, \dots, n\}$ . For each bidder  $i \in N$ , the strategy space is  $S_i = \mathbb{R}_+$ . For a bid profile  $s = (s_1, \dots, s_n) \in \prod_{i=1}^n S_i$ , define the highest bid by

$$M(s) = \max_{j \in N} s_j,$$

and the set of highest bidders by

$$H(s) = \{j \in N : s_j = M(s)\}.$$

The payoff of bidder  $i$  is

$$u_i(s) = \begin{cases} \frac{v_i - s_i}{|H(s)|}, & i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

- (b) **Second-price auction.** Again the player set is  $N = \{1, \dots, n\}$  and each bidder  $i$  chooses a bid  $s_i \in S_i = \mathbb{R}_+$ . Let

$$S(s) = \max\{s_j : j \in N \setminus H(s)\}$$

denote the highest bid among those who are not tied for the win. Then bidder  $i$ 's payoff can be written as

$$u_i(s) = \begin{cases} v_i - S(s), & \text{if } |H(s)| = 1 \text{ and } i \in H(s), \\ \frac{v_i - M(s)}{|H(s)|}, & \text{if } |H(s)| \geq 2 \text{ and } i \in H(s), \\ 0, & i \notin H(s). \end{cases}$$

## 2 Iterated Elimination

In the following normal-form game, which strategy profiles survive iterated elimination of strictly dominated strategies?

|          |     | Player 2 |      |      |
|----------|-----|----------|------|------|
|          |     | $L$      | $C$  | $R$  |
| Player 1 | $U$ | 6, 8     | 2, 6 | 8, 2 |
|          | $M$ | 8, 2     | 4, 4 | 9, 5 |
|          | $D$ | 8, 10    | 4, 6 | 6, 7 |

**Answer:** First, for Player 1, strategy  $U$  is strictly dominated by  $M$ , since

$$8 > 6, \quad 4 > 2, \quad 9 > 8.$$

Hence  $U$  is deleted.

After removing  $U$ , consider Player 2's payoffs in the reduced game with rows  $\{M, D\}$ . Now column  $C$  is strictly dominated by column  $R$ , because

$$5 > 4 \quad \text{and} \quad 7 > 6.$$

Hence  $C$  is deleted.

In the remaining game with rows  $\{M, D\}$  and columns  $\{L, R\}$ , no further strict dominance exists. Therefore, the strategy profiles that survive IESDS are

$$(M, L), \quad (M, R), \quad (D, L), \quad (D, R).$$

## 3 Roommates

Two roommates each need to choose to clean their apartment, and each can choose an amount of time  $t_i \geq 0$  to clean. If their choices are  $t_i$  and  $t_j$ , then player  $i$ 's payoff is given

by  $(10 - t_j)t_i - t_i^2$ . (This payoff function implies that the more one roommate cleans, the less valuable is cleaning for the other roommate.)

- (a) What is the best response correspondence of each player  $i$ ?
- (b) Which choices survive one round of IESDS?
- (c) Which choices survive IESDS?

**Answer:**

- (a) Fix player  $j$ 's choice  $t_j \geq 0$ . Player  $i$  solves

$$\max_{t_i \geq 0} u_i(t_i, t_j) = \max_{t_i \geq 0} ((10 - t_j)t_i - t_i^2).$$

Since this objective is strictly concave in  $t_i$ , the best response is unique. The first-order condition is

$$\frac{\partial u_i}{\partial t_i} = 10 - t_j - 2t_i = 0,$$

so the unconstrained maximizer is  $t_i = \frac{10 - t_j}{2}$ . Imposing  $t_i \geq 0$ , we obtain

$$BR_i(t_j) = \max \left\{ \frac{10 - t_j}{2}, 0 \right\}.$$

Equivalently,

$$BR_i(t_j) = \begin{cases} \frac{10 - t_j}{2}, & 0 \leq t_j \leq 10, \\ 0, & t_j \geq 10. \end{cases}$$

By symmetry, player  $j$  has the same best-response correspondence.

- (b) Let  $a, b \geq 0$  be two actions for player  $i$ . Then

$$u_i(a, t_j) - u_i(b, t_j) = (10 - t_j)a - a^2 - ((10 - t_j)b - b^2) = (a - b)(10 - t_j - a - b).$$

If  $b > 5$ , then action 5 strictly dominates  $b$ , because for every  $t_j \geq 0$ ,

$$u_i(5, t_j) - u_i(b, t_j) = (5 - b)(5 - t_j - b) > 0,$$

since  $5 - b < 0$  and  $5 - t_j - b < 0$  for all  $t_j \geq 0$  when  $b > 5$ . Hence every action  $b > 5$  is eliminated. On the other hand, no action in  $[0, 5]$  is strictly dominated in the original game. Indeed, if  $b \leq 5$  and some  $a > b$  were to strictly dominate  $b$ , we would need

$$(a - b)(10 - t_j - a - b) > 0 \quad \text{for all } t_j \geq 0,$$

which is impossible because for sufficiently large  $t_j$  the second factor becomes negative. If instead  $a < b$ , then strict dominance for all  $t_j \geq 0$  would require  $a + b > 10$ , which

cannot hold when  $b \leq 5$  and  $a < b$ . Therefore, after one round of IESDS, the surviving choices are exactly

$$[0, 5].$$

(c) Let the current surviving interval be  $[L, U]$ . For any  $a, b \in [L, U]$ ,

$$u_i(a, t_j) - u_i(b, t_j) = (a - b)(10 - t_j - a - b).$$

If opponents are restricted to  $t_j \in [L, U]$ , then every action below  $\frac{10-U}{2}$  is strictly dominated by  $\frac{10-U}{2}$ , and every action above  $\frac{10-L}{2}$  is strictly dominated by  $\frac{10-L}{2}$ . Hence one more round of elimination maps the interval  $[L, U]$  to

$$\left[ \frac{10-U}{2}, \frac{10-L}{2} \right].$$

Starting from the one-round survivor set  $[0, 5]$ , we obtain the sequence

$$[0, 5] \rightarrow \left[ \frac{5}{2}, 5 \right] \rightarrow \left[ \frac{5}{2}, \frac{15}{4} \right] \rightarrow \left[ \frac{25}{8}, \frac{15}{4} \right] \rightarrow \dots$$

These intervals converge to the unique fixed point  $t^*$  satisfying

$$t^* = \frac{10 - t^*}{2} \implies t^* = \frac{10}{3}.$$

Therefore, the only choice that survives IESDS is

$$t_i = \frac{10}{3}$$

for each player.

## 4 Campaigning

Two candidates, 1 and 2, are running for office. Each has one of three choices in running his campaign: focus on the positive aspects of one's own platform (call this a positive campaign  $[P]$ ), focus on the positive aspects of one's own platform while attacking one's opponent's campaign (call this a balanced campaign  $[B]$ ), or finally focus only on attacking one's opponent (call this a negative campaign  $[N]$ ). All a candidate cares about is the probability of winning, so assume that if a candidate expects to win with probability  $\pi \in [0, 1]$ , then his payoff is  $\pi$ . The probability that a candidate wins depends on his choice of campaign and his opponent's choice. The probabilities of winning are given as follows:

If both choose the same campaign then each wins with probability 0.5.

If candidate  $i$  uses a positive campaign while  $j \neq i$  uses a balanced one then  $i$  loses for sure.

If candidate  $i$  uses a positive campaign while  $j \neq i$  uses a negative one then  $i$  wins with probability 0.3.

If candidate  $i$  uses a negative campaign while  $j \neq i$  uses a balanced one then  $i$  wins with probability 0.6.

- (a) Model this story as a normal-form game. (It suffices to be specific about the payoff function of one player and to explain how the other player's payoff function is different and why.)
- (b) Write down the game in matrix form.
- (c) What happens at each stage of elimination of strictly dominated strategies? Will this procedure lead to a clear prediction?

**Answer:**

- (a) Let the set of players be  $N = \{1, 2\}$ , and each player's strategy set be

$$S_i = \{P, B, N\}, \quad i \in \{1, 2\}.$$

For player 1, the payoff function  $u_1 : S_1 \times S_2 \rightarrow [0, 1]$  is given by player 1's probability of winning:

$$u_1(P, P) = u_1(B, B) = u_1(N, N) = 0.5,$$

$$u_1(P, B) = 0, \quad u_1(B, P) = 1,$$

$$u_1(P, N) = 0.3, \quad u_1(N, P) = 0.7,$$

$$u_1(N, B) = 0.6, \quad u_1(B, N) = 0.4.$$

Since the game is symmetric and one candidate's probability of winning is the other candidate's probability of losing, player 2's payoff function is given by

$$u_2(s_1, s_2) = 1 - u_1(s_1, s_2).$$

- (b) The game in matrix form is therefore

|             |     | Candidate 2 |          |          |
|-------------|-----|-------------|----------|----------|
|             |     | $P$         | $B$      | $N$      |
| Candidate 1 | $P$ | 0.5, 0.5    | 0, 1     | 0.3, 0.7 |
|             | $B$ | 1, 0        | 0.5, 0.5 | 0.4, 0.6 |
|             | $N$ | 0.7, 0.3    | 0.6, 0.4 | 0.5, 0.5 |

- (c) First, for each player, strategy  $P$  is strictly dominated by  $N$ , so  $P$  is eliminated for both players. After deleting  $P$ , the reduced game has strategies  $\{B, N\}$  for each player. In this

reduced game, for each player,  $B$  is strictly dominated by  $N$ . Therefore  $B$  is eliminated as well, and the only surviving strategy for each player is  $N$ . Hence the unique strategy profile that survives IESDS is

$$(N, N).$$

So in this game, IESDS does lead to a clear prediction: both candidates run negative campaigns.

## 5 Beauty Contest Best Responses

Consider a game with  $n$  players, so that  $N = \{1, \dots, n\}$ . Each player has to choose an integer between 0 and 20, so that  $S_i = \{0, 1, 2, \dots, 19, 20\}$ . The winners are the players who choose an integer number that is closest to  $\frac{3}{4}$  of the average.

Because more than one person can win, define the set of winners  $W \subset N$  as those players who are closest to  $\frac{3}{4}$  of the average, and the rest are all losers. The set of winners is defined as

$$W = \left\{ i \in N : \arg \min_{i \in N} \left| s_i - \frac{3}{4} \frac{1}{n} \sum_{j=1}^n s_j \right| \right\}$$

To introduce payoffs, each player pays 1 to play the game, and winners split the pot equally among themselves. This implies that if there are  $m \geq 1$  winners, each gets a payoff of  $\frac{N-m}{m}$  (his share of the pot net of his own contribution) while losers get  $-1$  (they lose their contribution).

- Show that if player  $i$  believes that everyone else is choosing 20 then 19 is not the only best response for any number of players  $n$ .
- Show that the set of best-response strategies to everyone else choosing the number 20 depends on the number of players  $n$ .

**Answer:**

- Suppose every player other than  $i$  chooses 20, and let player  $i$  choose  $x \in \{0, 1, \dots, 20\}$ . Then the average is

$$\bar{s}(x) = \frac{20(n-1) + x}{n},$$

so the target number is

$$T(x) = \frac{3}{4} \bar{s}(x) = \frac{3}{4} \cdot \frac{20(n-1) + x}{n} = 15 - \frac{15}{n} + \frac{3x}{4n}.$$

Consider first  $x = 19$ . Then

$$T(19) = 15 - \frac{3}{4n},$$

so player  $i$ 's distance from the target is

$$|19 - T(19)| = 4 + \frac{3}{4n},$$

while each player choosing 20 is at distance

$$|20 - T(19)| = 5 + \frac{3}{4n}.$$

Hence 19 makes player  $i$  the unique winner.

Now consider  $x = 18$ . Then

$$T(18) = 15 - \frac{3}{2n},$$

and therefore

$$|18 - T(18)| = 3 + \frac{3}{2n}, \quad |20 - T(18)| = 5 + \frac{3}{2n}.$$

So 18 also makes player  $i$  the unique winner.

If player  $i$  is the unique winner, his payoff is  $n - 1$ . Thus both 18 and 19 yield payoff  $n - 1$ , so 19 is *not* the only best response for any  $n$ .

- (b) Again let player  $i$  choose  $x \in \{0, 1, \dots, 20\}$  while every other player chooses 20. If  $x = 20$ , then everyone ties as a winner, so player  $i$ 's payoff is

$$\frac{n - n}{n} = 0.$$

If  $x < 20$ , then the other  $n - 1$  players all choose the same number, so either player  $i$  is the unique winner or else he loses. Hence for  $x < 20$ ,

$$u_i(x, 20, \dots, 20) = \begin{cases} n - 1, & \text{if } |x - T(x)| < |20 - T(x)|, \\ -1, & \text{if } |x - T(x)| > |20 - T(x)|. \end{cases}$$

Therefore, the best responses are exactly those  $x < 20$  for which player  $i$  is strictly closer to the target than 20 is.

We solve

$$|x - T(x)| < |20 - T(x)|.$$

Since  $x < 20$ , we have  $x - 20 < 0$ , so after squaring both sides and factoring,

$$(x - T(x))^2 < (20 - T(x))^2 \iff (x - 20)(x + 20 - 2T(x)) < 0 \iff x + 20 - 2T(x) > 0.$$

Substituting for  $T(x)$  gives

$$x + 20 - 2 \left( 15 - \frac{15}{n} + \frac{3x}{4n} \right) > 0,$$

which simplifies to

$$x \left( 1 - \frac{3}{2n} \right) > 10 - \frac{30}{n} \iff x(2n - 3) > 20(n - 3).$$

Thus the set of best responses is

$$BR_i(20, \dots, 20) = \left\{ x \in \{0, 1, \dots, 19\} : x > \frac{20(n - 3)}{2n - 3} \right\}.$$

This set depends on  $n$ . For example,

$$BR_i(20) = \{0, 1, \dots, 19\} \quad \text{when } n = 2,$$

while

$$BR_i(20, 20, 20) = \{5, 6, \dots, 19\} \quad \text{when } n = 4.$$

Hence the set of best-response strategies to everyone else choosing 20 does indeed depend on the number of players  $n$ .

## 6 IESDS and BR in Infinite Games

Consider the following two-player game. Each player announces a nonnegative real number. The payoffs are

$$v_i(x_i, x_j) = \begin{cases} 2, & \text{if } x_i = 0 \text{ and } x_j = 1, \\ \arctan x_i, & \text{if otherwise.} \end{cases}$$

- Argue that every positive announcement is strictly dominated.
- Argue that announcement 0 is not strictly dominated.
- From the above two questions, we know only 0 survives IESDS for both players. Are they mutual best responses?

**Answer:**

- Fix any announcement  $x_i > 0$ . We claim that  $x_i$  is strictly dominated by any larger announcement  $y_i > x_i$ . Indeed, for every announcement  $x_j \geq 0$  of player  $j$ , if player  $i$  chooses  $x_i > 0$ , then the special case in the payoff definition never applies, so

$$v_i(x_i, x_j) = \arctan x_i.$$



Likewise,

$$v_i(y_i, x_j) = \arctan y_i.$$

Since  $\arctan(\cdot)$  is strictly increasing on  $\mathbb{R}_+$  and  $y_i > x_i$ , we have

$$\arctan y_i > \arctan x_i.$$

Therefore,

$$v_i(y_i, x_j) > v_i(x_i, x_j) \quad \text{for all } x_j \geq 0,$$

so every positive announcement is strictly dominated.

- (b) Let  $y_i \geq 0$  be any alternative announcement. If  $y_i = 0$ , then  $y_i$  clearly does not strictly dominate 0, since the two strategies are identical.

If  $y_i > 0$ , consider the opponent announcement  $x_j = 1$ . Then

$$v_i(0, 1) = 2,$$

while

$$v_i(y_i, 1) = \arctan y_i < \frac{\pi}{2} < 2.$$

Hence  $y_i$  cannot strictly dominate 0. Since no strategy  $y_i \geq 0$  strictly dominates 0, announcement 0 is not strictly dominated.

- (c) From parts (a) and (b), the only strategy that survives IESDS is 0 for each player. However, these are *not* mutual best responses.

To see this, fix  $x_j = 0$ . Then for player  $i$ ,

$$v_i(x_i, 0) = \arctan x_i \quad \text{for all } x_i \geq 0,$$

because the exceptional payoff 2 occurs only when  $(x_i, x_j) = (0, 1)$ . Since  $\arctan x_i$  is strictly increasing in  $x_i$ , we have

$$\arctan x_i > \arctan 0 = 0 \quad \text{for every } x_i > 0.$$

So  $x_i = 0$  is not a best response to  $x_j = 0$ .

In fact, there is no best response to  $x_j = 0$  at all: the payoff  $\arctan x_i$  can be made arbitrarily close to  $\pi/2$  by choosing larger and larger  $x_i$ , but it never reaches  $\pi/2$  for any finite  $x_i$ . Thus the best-response set to 0 is empty.

Therefore, although IESDS leaves only the strategy profile  $(0, 0)$ , that profile does not consist of mutual best responses.

## 7 $n$ -Firm Cournot Competition

Consider the  $n$ -firm Cournot competition. The demand curve is still

$$D(Q) = \max\{100 - Q, 0\}.$$

If each firm  $i$  supplies  $q_i$ , the total supply is  $\sum_{i=1}^n q_i$ . Suppose each firm's marginal cost is 10.

- (a) Write down its normal form game.
- (b) For each firm, what are the strategies that survive IESDS?

**Answer:**

- (a) Let the set of firms be  $N = \{1, \dots, n\}$ . Firm  $i$  chooses an output level  $q_i \in S_i = \mathbb{R}_+$ . Since each firm has constant marginal cost 10, firm  $i$ 's profit is

$$v_i(q_1, \dots, q_n) = q_i(D(Q) - 10) = q_i \left( \max \left\{ 100 - \sum_{k=1}^n q_k, 0 \right\} - 10 \right).$$

Equivalently,

$$v_i(q) = \begin{cases} q_i (90 - \sum_{k=1}^n q_k), & \text{if } \sum_{k=1}^n q_k \leq 100, \\ -10q_i, & \text{if } \sum_{k=1}^n q_k > 100. \end{cases}$$

Therefore the normal-form game is

$$G = (N, (S_i)_{i \in N}, (v_i)_{i \in N})$$

with  $N = \{1, \dots, n\}$ ,  $S_i = \mathbb{R}_+$  for every  $i$ , and payoff functions  $v_i$  as above.

- (b) We first derive firm  $i$ 's best response to the total output of the other firms. Let

$$x = \sum_{j \neq i} q_j.$$

Then firm  $i$  chooses  $q_i \geq 0$  to maximize

$$q_i(\max\{100 - x - q_i, 0\} - 10).$$

If  $x \geq 90$ , every positive  $q_i$  yields negative profit, so the best response is  $q_i = 0$ . If  $x < 90$ , then the relevant objective is

$$q_i(90 - x - q_i),$$

whose unique maximizer is obtained from the first-order condition

$$90 - x - 2q_i = 0.$$

Thus

$$BR_i(x) = \max \left\{ \frac{90 - x}{2}, 0 \right\}.$$

We now perform IESDS.

First, every quantity  $q_i > 90$  is strictly dominated by 0, because for every profile of the other firms,

$$v_i(q_i, q_{-i}) < 0 = v_i(0, q_{-i}).$$

Hence after the first round each firm's surviving strategies lie in

$$[0, 90].$$

Next, in this reduced game every quantity  $q_i > 45$  is strictly dominated by 45. Indeed, let  $x = \sum_{j \neq i} q_j \geq 0$  and take any  $q_i > 45$ . If  $x + q_i \leq 100$ , then both profits are computed in the positive-price region, and

$$v_i(45, x) - v_i(q_i, x) = (45 - q_i)(45 - x - q_i) > 0,$$

because  $45 - q_i < 0$  and  $45 - x - q_i < 0$ . If  $x + q_i > 100$ , then  $v_i(q_i, x) = -10q_i < -450$ . On the other hand, if  $x \leq 55$ , then

$$v_i(45, x) = 45(45 - x) \geq -450,$$

while if  $x > 55$ , then also  $x + 45 > 100$ , so

$$v_i(45, x) = -450.$$

In either case  $v_i(45, x) > v_i(q_i, x)$ . Hence every quantity above 45 is eliminated. Therefore, after two rounds, each firm's surviving strategies are contained in

$$[0, 45].$$

At this point the answer depends on  $n$ .

If  $n = 2$ , let the current surviving interval be  $[L, U] \subseteq [0, 45]$ . Since the opponent's output lies in  $[L, U]$ , firm  $i$ 's best responses lie in

$$\left[ \frac{90 - U}{2}, \frac{90 - L}{2} \right].$$

Hence one more round of elimination maps  $[L, U]$  to

$$\left[ \frac{90 - U}{2}, \frac{90 - L}{2} \right].$$

Starting from  $[0, 45]$ , we obtain

$$[0, 45] \rightarrow \left[ \frac{45}{2}, 45 \right] \rightarrow \left[ \frac{45}{2}, \frac{135}{4} \right] \rightarrow \dots$$

and these intervals converge to the unique fixed point  $q^*$  satisfying

$$q^* = \frac{90 - q^*}{2}.$$

Solving gives

$$q^* = 30.$$

So when  $n = 2$ , the only strategy that survives IESDS is

$$q_i = 30.$$

If  $n \geq 3$ , no further elimination is possible after reaching  $[0, 45]$ . To see this, fix any  $q \in [0, 45]$ . Then

$$x = 90 - 2q \in [0, 90].$$

Because there are at least two rival firms, we can choose their outputs in  $[0, 45]$  so that their total output is exactly  $x$  (for instance, let one rival produce  $\min\{45, x\}$ , another produce the remainder, and any further rivals produce 0). Against such a profile, firm  $i$ 's best response is

$$BR_i(x) = \frac{90 - x}{2} = q.$$

Hence every  $q \in [0, 45]$  is a best response to some surviving profile of the other firms, so no such strategy can be strictly dominated. Therefore, when  $n \geq 3$ , the strategies that survive IESDS are exactly

$$[0, 45].$$